Introduction

Evolutionary rescue is a central concept in eco-evolutionary dynamics, describing how populations can avoid extinction through adaptive evolution following abrupt environmental change. In a theoretical framework, Martin et al. (2013) showed that the probability of evolutionary rescue (PR) increases with initial population size (N_0) according to the following relationship:

$$\boldsymbol{P_R = 1 - \exp\left(-\theta N_0\right)}$$

where θ is a composite parameter typically containing mutation rate and/or frequency of pre-existing mutants and the temporal dynamics of wild-type population size. Critically, θ is assumed independent of N_0. This positive relationship is now widely regarded as intuitive and widely adopted in theoretical models.

Despite this widespread acceptance, the empirical foundations of this relationship remain surprisingly weak. Indeed, the link between N_0 and PR is frequently assumed rather than evaluated. This gap between theory and evidence raises a simple yet fundamental question: how well supported by data is the presumed relationship between initial population size and the probability of evolutionary rescue?

Methods

To address this question, we conducted a systematic examination of the literature citing Martin et al. (2013). Citations were retrieved using Web of Knowledge, with the primary search conducted in November 2025. In total, 56 citations of the Martin et al. (2013) paper (PMID: 23209169) were identified on PubMed (retrieved on October 16, 2025).

To provide a broader and more exhaustive overview of empirical work on evolutionary rescue, we expanded our search to include citations of two additional landmark studies: Bell and Gonzalez (2009) and Ramsayer et al. (2013). Citation lists for these papers were retrieved from PubMed on November 12, 2025. The Bell and Gonzalez (2009) paper yielded 176 citations, while the Ramsayer et al. (2013) paper yielded 44 citations. We accounted for overlap among citation lists by identifying papers that cited more than one of the three focal studies, resulting in 38 duplicated papers. After eliminating duplicates, we screened a total of 229 unique papers.

We restricted our analysis to studies of evolutionary rescue, defined as population persistence driven by adaptive evolution following an environmentally induced population decline, as opposed to adaptation without demographic collapse.

Results

Bell & Gonzalez (2009), whose data was re-analysed by Martin et al. (2013), empirically tested the relationship between initial population size (N_0) and the probability of evolutionary rescue in the yeast *Saccharomyces cerevisiae* across 10 initial population sizes. Another early study by Samani and Bell (2010) in the yeast *Saccharomyces paradoxus* also used a wide range of initial sizes, but they did not quantify the probability of evolutionary rescue, as their experimental design involved gradually increased salt concentrations until populations showed a substantial reduction in density, but were not extinct. Among the remaining 229 papers examined, only a handful tested the relationship between probability of evolutionary rescue and initial population size, and these studies did not use a broad gradient of population sizes. Rather, they relied exclusively on comparisons between only two population sizes.

For instance, Ramsayer et al. (2013) showed that population size and genetic diversity affected the probability of evolutionary rescue in the bacterium *Pseudomonas fluorescens*, but population size was contrasted only between large and small treatments. Similar binary designs were used in several studies with flour beetles (*Tribolium castaneum*): Hufbauer et al. (2015) compared populations founded with 50 versus 150 individuals, Stewart et al. (2017) used founder sizes of 10 versus 40 individuals, and Lewis et al. (2024) contrasted populations of 100 individuals with scaled-up populations of approximately 10,000 individuals, all reporting higher extinction risk or lower fitness in smaller populations.

Thus, while population size is often included as a factor in experimental designs, it is rarely treated as a focal variable whose quantitative relationship with rescue probability is explicitly tested.

Additionally, it is important to separate the terms of “population size” and “density”. Population size is defined as the absolute number of individuals (*N*) in a population at a given timepoint. In contrast, the density of a population is defined as the number of individuals per unit of space (e.g. per m^2) or in relation to the carrying capacity, and a high population density can limit growth due to competition. However, there is no strict separation of population size and density in evolutionary rescue literature and variable experimental designs were used: Bell and Gonzalez (2009) used 10 different initial densities in the same culture volume/unit of space, implying that initial population size and density co-varied. In contrast, both Samani and Bell (2010) and Ramsayer et al. (2013) used a fixed initial density across multiple different culture volumes to achieve different initial population sizes. Olazcuaga et al. (2026) investigated density dependence in *Tribolium castaneum* and compared populations with low and constant

density. By comparing populations with the same initial size, they showed that populations evolving with negative density dependence (i.e., habitat of the same size throughout the experiment) had a lower probability of rescue than populations evolving without density dependence (i.e., with habitat that varies with population size, maintaining a constant density).

The aforementioned studies all considered evolutionary rescue in the simplest case of a non-spatially-structured, single-species population in a single-step environmental change. A few experimental studies (amongst the screened citations or other papers known to us) have varied initial population size, or its analogue, in more complex experimental designs. For instance, Zhang et al. (2011) looked at emergence of antibiotic resistance in the bacterium *E. coli* on a spatially structured microfluidic culture device inoculated with varying population sizes. However, as no extinction was detected, it is debatable whether this study observed evolutionary rescue. Vogwill et al. (2016) treated populations of *P. fluorescens* with an antibiotic concentration that still allowed some population growth (80% of MIC), but passaged populations with varying bottleneck sizes, such that populations showed net decline and some went extinct while others were rescued. Meanwhile, studies of multi-species community rescue (Low-Décarie et al. 2015; Fugère et al. 2020) have varied initial biomass of the entire community. Although all of these studies provide interesting qualitative results on the role of initial population/community size, the corresponding theory is not sufficiently well developed for these more complex scenarios to provide a quantitative test of predictions.

Discussion

Our review reveals a striking disconnect between theory and empirical practice. We identify three major limitations that characterize the existing empirical literature. First, most studies rely on only two values of N_0, precluding any assessment of non-linearities, thresholds, or diminishing returns in rescue probability. Second, population size is often embedded within complex experimental designs and treated as a background parameter rather than a variable of primary interest. As a result, its effect on rescue probability is frequently intertwined with other ecological or evolutionary processes, making it difficult to isolate the independent contribution of N_0. Thirdly, population size (absolute number of individuals) and density (individuals per unit area/volume) have not been clearly distinguished. Some studies have used fixed density and varying volume to achieve different initial population sizes, while others have confounded population size and density by using a fixed volume.

Together, these limitations highlight the need for experimental studies explicitly designed to test how evolutionary rescue probability scales with initial population size. Addressing this gap is essential for grounding theoretical predictions in empirical evidence

and for improving our ability to forecast population persistence under rapid environmental change.

Brainstorming part on perspective

Perspective: How would we like to design experiments in the future? Give a couple of examples of experimental systems and how we would **separately manipulate abundance/density**? E.g. bacteria, flour beetles… → everyone brainstorm

We can vary absolute cell numbers while keeping population density constant. And we can vary population density while keeping absolute cell numbers constant.

- What is available/needed for further experimental testing?

[Helen]
Here are some experimental studies that are not testing evolutionary rescue, but have used some relevant design choices:
Bacteria:

- Alexander and MacLean (2020) separated effects of population size and density in bacteria by varying culture volumes, focussing on determining minimal inhibitory concentrations of a *P. aeruginosa* strains with different inoculum sizes.
- Chavhan et al. (2019), with *E. coli*, achieve fixed initial culture density with two different culture volumes (1.5ml vs. 100ml).
- Schenk et al. (2022), with *E. coli*, also use two different culture volumes to test evolution of antibiotic resistance from two different effective population sizes.

Insects:

- Willi & Hoffmann (2009) study experimental populations of *Drosophila birchii* in three different initial population sizes (20, 100, or 1000 individuals) but control for population density by dividing the larger populations across multiple vials.

[Helen] – In our lab we are currently doing pilot experiments disentangling population size and density in different culture volumes, using bacterial (*P. aeruginosa*) populations. Currently we are looking at demographic rather than evolutionary responses (although we can never entirely rule out evolution during the course of an experiment). We mostly do experiments on microtitre (multi-well) plates – here's the range of feasible culture volumes:

- 6-well: max 2ml recommended for plate reader; unsure of minimum (we have not tested ourselves)
- 12-well: we manage to culture up to 3ml, but the plate manufacturer recommends a range of only 0.76-1.14 ml and the plate reader recommends max 1.2ml
- 24-well: we manage a range of 0.4-1ml
- 48-well: max 0.4ml recommended for plate reader (we haven't tested ourselves)
- 96-well: max 200ul. Not entirely sure of min but I think around 100ul.
- 384-well, etc: Available, but I probably wouldn't culture in these.

Of course, it is also possible to use culture tubes and flasks to achieve much larger volumes – they're just more laborious for high replication.

Some considerations:

- Different plate formats / culture vessels differ in more than just the volume of culture. For example, surface area to volume ratio, head-room in the vessel, type of cap/seal, and degree of mixing in different geometries affect oxygenation of the culture. Therefore, there are subtly different environments that could lead to different growth rates if the same culture volume & density is placed in different well/vessel formats.
- The range of volumes achievable in any given plate format is rather limited (up to ~2-3 fold on microtitre plates), which limits the ability to control for potential confounders of using different plate formats.
- Population density is more readily varied over orders of magnitude. For instance, a saturated culture of *P. aeruginosa* in rich liquid media has a density of ~5 x 10^9 CFU/ml and we can dilute that as far as we want. (We would normally dilute at least 10-fold from a pre-culture to start an antibiotic treatment, but it's also possible to spin down to concentrate cells.)
- All of the above is assuming you culture in liquid media (broth), which would be most readily comparable to existing theory for non-spatially-structured populations. However, mixing in culture vessels is not perfect, so it's not precisely true that populations are well-mixed/spatially homogeneous (e.g. cells may aggregate into clumps, stick to the walls of culture vessels, etc.). This could be a source of small discrepancies from theory, but I don't think it should prevent going ahead and testing.
- You can also culture bacteria on solid media (agar), but then it is less reasonable to ignore spatial structure.